

Wide range linearization calibration method for DFB Laser in FMCW LiDAR

Jiatong Zhang, Chang Liu, Liwen Su, Xinghu Fu, Wa Jin, Weihong Bi, Guangwei Fu^{*}

School of Information Science and Engineering, The Key Laboratory for Special Fiber and Fiber Sensor of Hebei Province, Yanshan University, Qinhuangdao 066004, China

ARTICLE INFO

Keywords:

Frequency-modulated continuous-wave (FMCW) LiDAR
Phase-locked loop (PLL)
DFB laser

ABSTRACT

In this paper, a solution based on a hybrid electro-optic phase-locked loop (HEO-PLL) is proposed to address the nonlinearity of the continuous sweep frequency of the light source in Frequency Modulated Continuous Wave (FMCW) LiDAR. This nonlinearity leads to large measurement errors, low accuracy, and poor system adaptability. The proposed solution uses a HEO-PLL to control the frequency sweep linearity of the distributed feedback (DFB) laser when it undergoes continuous, rapid, and wide-range frequency modulation through current tuning. The system is designed to allow the DFB laser to be used as a high-quality source for FMCW LiDAR. This enhances the resolution and ranging capability of the FMCW LiDAR with the DFB laser as the source, while avoiding issues such as difficult tuning of the loop bandwidth and central frequency and being highly susceptible to temperature and voltage variations caused by the charge pump phase-locked loop.

1. Introduction

Frequency Modulated Continuous Wave (FMCW) LiDAR is a type of LiDAR technology that uses a continuous wave laser with a modulated frequency to detect and measure distances to objects [1–4]. It works by emitting a laser beam with a continuously varying frequency (i.e., a chirp), and measuring the time delay between the transmitted and reflected signals to convert the beat signal to determine the distance to a target. The principle of FMCW LiDAR is based on the Doppler effect, which causes a change in frequency of the reflected signal as a result of the target's motion. By analyzing the frequency shift of the beat signal, FMCW LiDAR can also provide information on the velocity of the target [5–8]. FMCW LiDAR has several advantages over other LiDAR technologies, including its high range resolution, accuracy, and ability to measure both range and velocity. It is commonly used in applications such as autonomous vehicles [9], robotics [10], and environmental monitoring [11,12].

In FMCW LiDAR, there are two ways to generate sweep frequency modulated laser. The first method is external modulation [13,14], which uses a Electro-optic modulator to load a sweep electrical signal onto a fixed-frequency laser, typically using a high-bandwidth microwave source for arbitrary signal generator to generate the sweep electrical signal, however, high-bandwidth, high-linearity microwave sources are extremely expensive, and their sweep range is limited. Another method is the internal modulation of the laser, by changing the driving

conditions of the laser to change the frequency of the output laser. The most direct is the use of the distributed feedback (DFB) laser, whose output frequency is dependent on its temperature and current [15,16], the chirped laser is produced by modulating the injection current or the control temperature. However, in the application of current-modulated DFB laser, the relationship between the injection current and the output frequency is not linear, so it is necessary to correct the output laser. One of the common methods used to correct the sweep nonlinearity of DFB lasers is to modulate the laser with a prefabricated current by processing the frequency error with an algorithm and correcting the injection current signal [17], but it is difficult to deal with the influence of the external environment or the temperature of the laser itself. Another strategy is to use hardware to build optical phase-locked loop (OPLL) [18–20], which can provide active real-time feedback to the laser, it is difficult to realize the linear frequency sweep of laser, but the system structure of this method is complex, difficult to debug, difficult to transplant the system and so on.

To sum up, in order to deal with these challenges, this paper built a DFB laser frequency scanning nonlinear correction system based on FPGA, the core of the system is the combination of open-loop pre-correction and hybrid electro-optic phase-locked loop (HEO-PLL). The digital part including triangle wave modulation signal and error correction are generated and implemented in FPGA. The analog Voltage-controlled oscillator consists of a DFB laser, an auxiliary unbalanced Mach-Zehnder interferometer (MZI), and a light balance detector. By

^{*} Corresponding author.

E-mail address: earl@ysu.edu.cn (G. Fu).

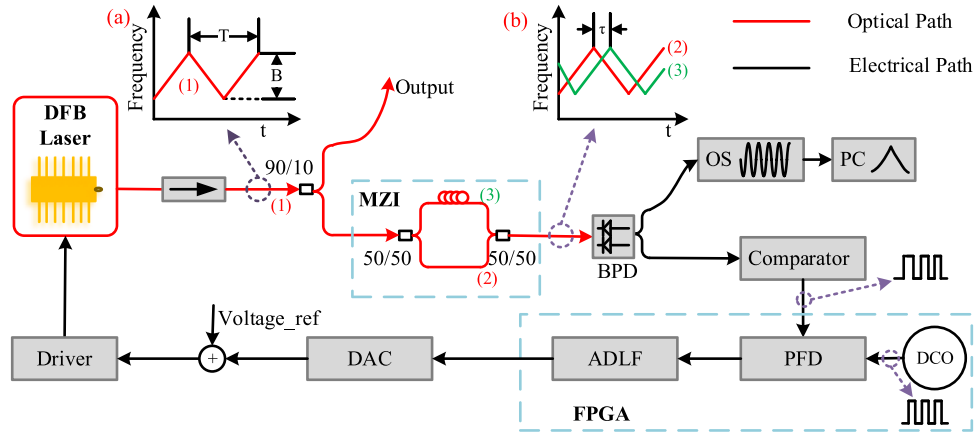


Fig. 1. HEO-PLL correction system. DFB Laser: Distributed feedback laser; MZI: Mach-Zehnder interferometer; BPD: Balance Photoelectric Detector; OS: Oscilloscope; PC: Personal computer; FPGA: Field Programmable Gate Array; DCO: Digital controlled oscillator; PFD: Phase frequency detector; ADLF: Adaptive Loop filter; DAC: Digital to analog converter.

combining current tuning and electrical feedback compensation, the DFB laser has a tuning period of 10 ms and a tuning range of 67.8 GHz with high coherence, precision and linear sweep frequency.

2. Principle

Fig. 1 shows a HEO-PLL system designed for wide-range, linear-frequency scanning, featuring an open-loop soft correction mechanism based on an iterative algorithm. In the open-loop soft correction phase, the DFB laser undergoes initial driving via a standard triangle wave current signal. The oscilloscope records the beat signal from the balanced photoelectric detector (BPD), and the beat signal undergoes a Hilbert transformation in the computer. By analyzing the time-frequency curve of the beat signal and considering the relationship between the laser current and output laser frequency, a pre-corrected driving signal is derived. This pre-corrected driving signal is then input into the FPGA, leveraging the FPGA to output a direct digital synthesizer (DDS) signal that drives the DFB laser. This process iterates until the beat signal fluctuations and bandwidth reach a minimum.

In the system's configuration, **Fig. 1(a)** depicts the time-frequency curve of the periodic sweep laser signal emitted by the DFB laser. Following the laser signal's passage through the optical isolator, a 90/10 coupler divides it into two beams: ninety percent serves as a direct detection signal, while the remaining 10 percent enters the unbalanced MZI as a correction signal. **Fig. 1(b)** illustrates the time-frequency curves of the two laser signals with a delay τ . The BPD converts these two coherent laser signals into a beat signal in the form of an electrical signal. Subsequently, the beat signal is transformed into a TTL square wave signal, encompassing both frequency and phase information, after passing through a high-speed comparator. Upon entering the FPGA, this square wave signal, along with the square wave signal generated by the internal digital controlled oscillator (DCO), engages in digital operations within the phase frequency detector (PFD). The PFD produces a phase error sequence, which, after adaptive filtering, is superimposed with the pre-corrected signal waveform. The FPGA's DDS module then acts on the laser driver in tandem with the bias voltage signal. In the closed-loop system, the phase error between the beat signal and the square wave signal generated by the DCO converges to zero, indicating that the output of the DFB laser manifests as a linear sweep laser signal.

Suppose the instantaneous frequency of the laser emission can be expressed as [21]

$$f_{\text{out}} = f_0 + i(t)K[i(t)] \quad (1)$$

Where f_0 is the initial frequency of the laser, $i(t)$ is the driving current of the laser, $K[i(t)]$ is the sweep frequency of the laser, according to the

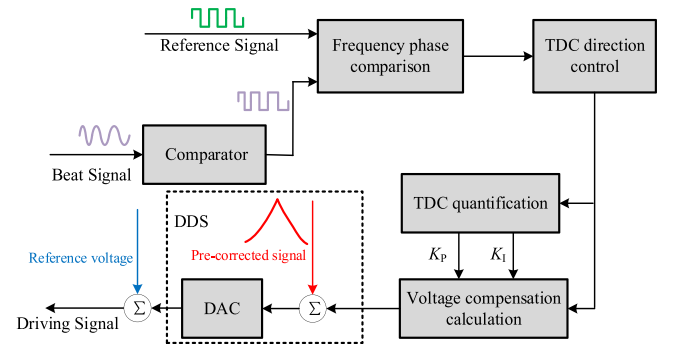


Fig. 2. Schematic diagram of ADLF.

sweep characteristics of DFB laser and the characteristics of unbalanced MZI, the condition of linear sweep laser output of DFB laser can be expressed as

$$f_{\text{beat}} = k\tau = \tau \frac{df_{\text{out}}}{dt} = \frac{di(t)}{dt} \left\{ \tau \frac{dK[i(t)]}{dt} i(t) + \tau K[i(t)] \right\} \quad (2)$$

In the process of DFB laser soft correction, the DFB laser is driven by a triangle wave with a fixed slope, and the interference signal is generated by the unbalanced MZI. By testing the frequency modulation characteristics of the laser, the ideal beat frequency of the DFB laser can be predicted when the length of the reference arm of the unbalanced MZI is fixed, according to **Eq. (2)**, the nonlinear tunability of the laser can be obtained by using the time-frequency information of the beat signal and the ideal beat frequency; After determining the non-linear tunable quantity of the laser, the triangle wave driving signal is reconstructed according to **Eq. (2)**, and the laser is driven to adjust the frequency again.

In the HEO-PLL correction part, the high-speed comparator is used as the connecting device between the BPD and FPGA, on the one hand, it can extract the frequency and phase information of the beat frequency signal, on the other hand, it avoids the influence of amplitude modulation in the process of laser frequency modulation.

The schematic diagram of adaptive loop filter (ADLF) involved in the driving signal of DFB laser is shown in **Fig. 2**. The PFD inside the FPGA uses bidirectional time-to-digital converter (TDC) [22] to quantify the error signal, it can realize the quantization of the phase error of the beat signal in different sweep direction of DFB laser. ADLF adopts adaptive control based on proportional integral (PI), and the gain coefficients K_P and K_I of PI are self-regulated with the magnitude of TDC quantization phase error. The beat signal keeps good stability under static and

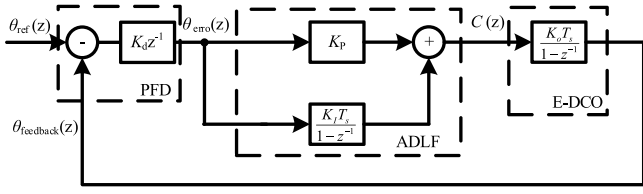


Fig. 3. The Z-domain model of HEO-PLL constructed.

dynamic conditions. This method can provide smaller phase compensation when the beat signal produces weaker phase error and larger phase compensation when it produces stronger phase error. On the one hand, it realizes the same frequency locking effect of the beat signal, on the other hand, it speeds up the locking rate of the HEO-PLL and improves the stability of the HEO-PLL. In the experiment, we fix other parameters, continuously adjusted the value of gain coefficient of K_P , K_I , and observe the beat signal time-frequency curve to determine the best gain coefficient.

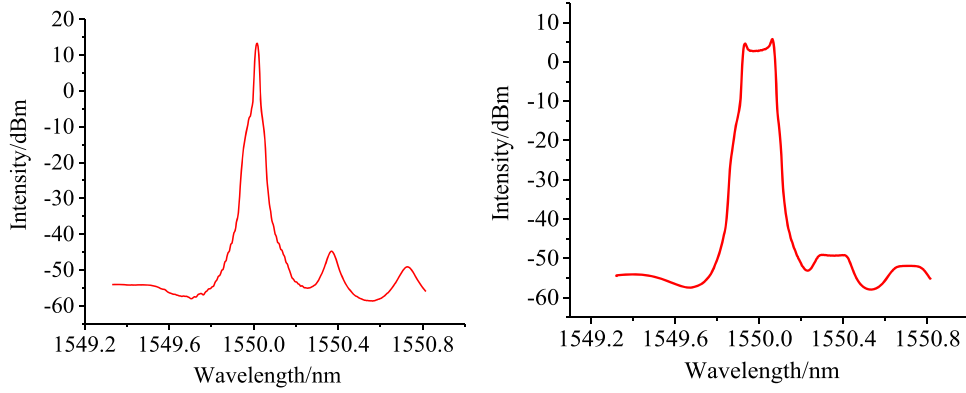


Fig. 4. Spectra of the DFB laser. (a) static output. (b) dynamic output.

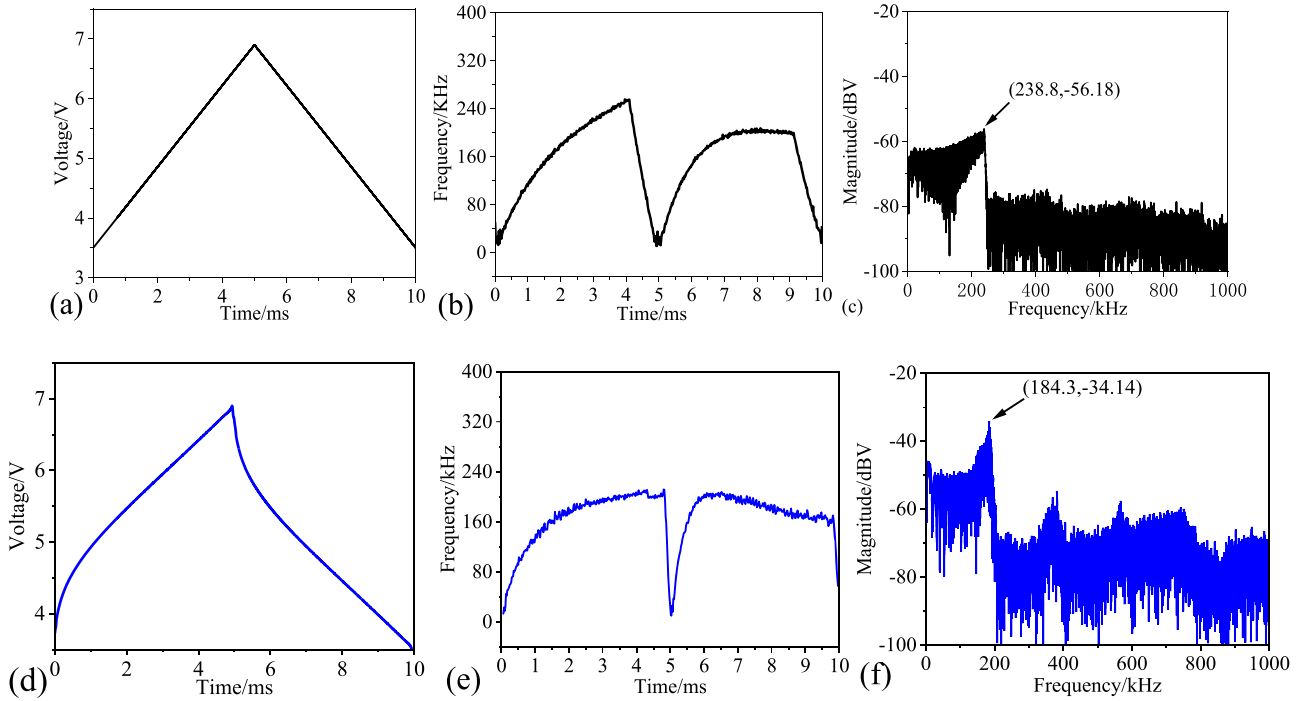


Fig. 5. Frequency sweep experiment of DFB laser linearization. (a)-(c) The driving signal, the time-frequency curve of the beat signal and the FFT spectrum of the beat signal in the DFB laser no-correction experiment. (d)-(f) The driving signal, the time-frequency curve of the beat signal and the FFT spectrum of the beat signal in the open-loop soft correction experiment of DFB laser.

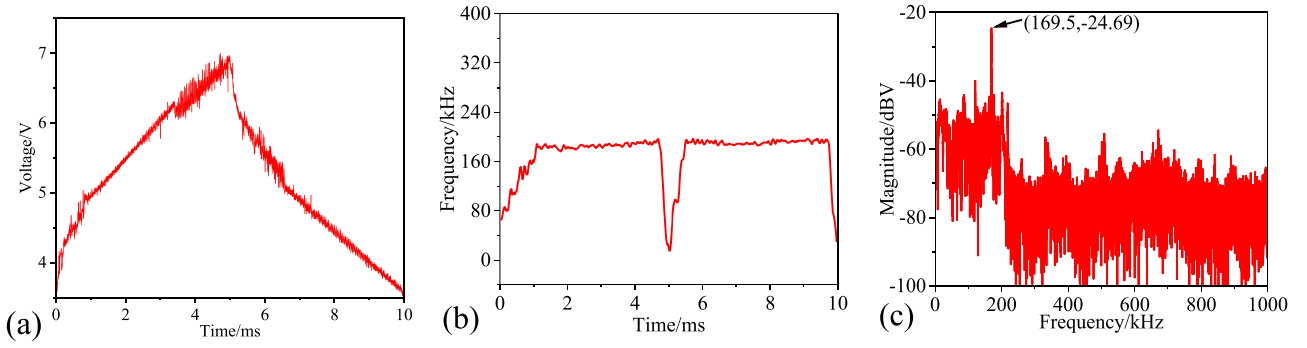


Fig. 6. Frequency-sweep linearization experiment of DFB laser based on HEO-PLL. (a) The driving signal; (b) The time-frequency curve of the beat signal; (c) The FFT spectrum of the beat signal.

3. Experimental setup and results

In the experiment, a commercial narrow linewidth center wavelength tunable DFB laser with a linewidth of 380 kHz, a maximum output power of 40 mW, and a side mode suppression ratio of 57 dB at a center wavelength of 1550.108 nm was used. The static and dynamic output spectra of the laser are shown in Fig. 4.

The delay fiber length of the unbalanced MZI is 5 m, which is moderate and ensures that the laser coherence condition is met while the phase error caused by the nonlinear chirp of the laser can be corrected by the loop. In the experiment, we found that when analyzing the linearity of the DFB laser sweep using only the Hilbert transform, due to the complexity of the beat frequency signal frequency, the selectivity problem of the region of interest (ROI) [20,23,24] of the beat frequency signal, etc., the trend of the transform was judged using the time-frequency curve of the beat frequency signal, and the 3 dB bandwidth of the beat frequency signal was used to analyze the linearity of the laser sweep. The frequency spectrum or time-frequency curve of the beat frequency signal can be measured to characterize the linearity of the laser sweep signal, and the actual laser sweep curve can also be derived by Hilbert transforming the beat frequency signal. As shown in Fig. 1, the experiment uses an oscilloscope to obtain the beat frequency signal, which can obtain the FFT spectrum of the beat frequency signal, and the time-frequency curve of the beat frequency signal can be obtained by using the Hilbert transform. The DFB laser sweep period is 10 ms, and its drive signal includes a bias voltage of 5.2 V provided by a DC voltage source and a pre-calibrated triangular wave voltage of ± 1.7 V provided by a DAC. The drive signals are controlled and output by an FPGA.

3.1. Open-loop soft correction experiment

During the experiment, the initial triangular wave drive voltage

signal of the DFB laser is shown in Fig. 5(a), and the time-frequency curve of the beat frequency signal output from the unbalanced MZI is shown in Fig. 5(b). In the Fig. 5(b), it can be found that the time-frequency curve is not constant within a single chirp cycle, but the instantaneous frequency of the beat frequency signal changes slowly with time. The FFT power spectrum of Fig. 5(c) shows that the center frequency of the beat frequency signal is 238.8 kHz, and its 3 dB bandwidth is 39.5376 kHz. It can be calculated that the nonlinearity of the laser sweep under the initial triangular wave signal drive is 16.55 %. This paper uses the open-loop soft correction method. After the fourth iteration, the nonlinearity of the DFB laser reaches its minimum and then continues to deteriorate. The drive signal of the DFB laser obtained after the fourth iteration is shown in Fig. 5(d), and the time-frequency curve of its beat frequency signal is analyzed in Fig. 5(e). Compared with the initial triangular wave drive, the amplitude of the instantaneous frequency of the beat frequency signal decreases with time. The FFT power spectrum of Fig. 5(f) shows that the center frequency of the beat frequency signal is 174 kHz, and its 3 dB bandwidth decreases to 6 kHz. It can be calculated that under the open-loop soft correction, the nonlinearity of the laser sweep is reduced to 3.45 %.

3.2. HEO-PLL calibration experiments

When HEO-PLL is introduced into the calibration system of DFB laser, the driving signal, the time-frequency curve and FFT spectrum of its beat signal are shown in Fig. 5.

After adding the HEO-PLL structure, the driving signal of the DFB laser was as shown in Fig. 6(a), and the fluctuation of the driving signal means that the HEO-PLL worked continuously and changed dramatically during the course of the direction shift. By analyzing the information brought by Fig. 6(b) and Fig. 6(c), the fluctuation of the beat signal in the sweep cycle is further reduced, about 80 % of the frequency is locked at 169.5 kHz, and the power of the beat signal is increased to -24.69 db.

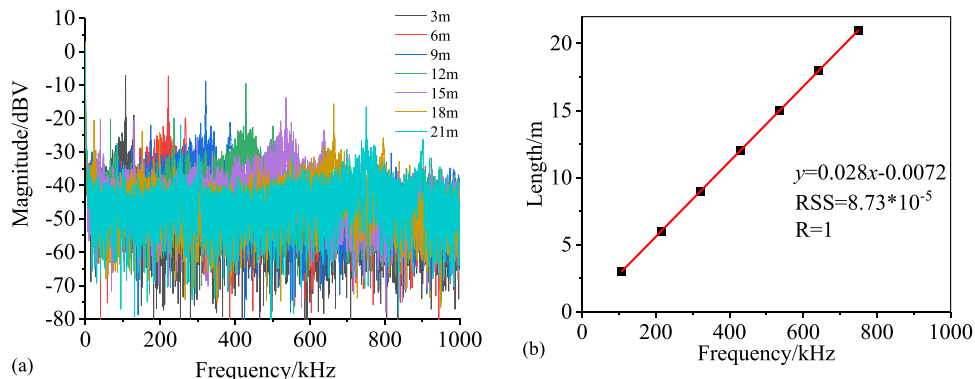


Fig. 7. (a) FFT spectrum of beat signal in 3m-21m ranging experiment. (b) Distance fitting curve of distance measurement experiment.

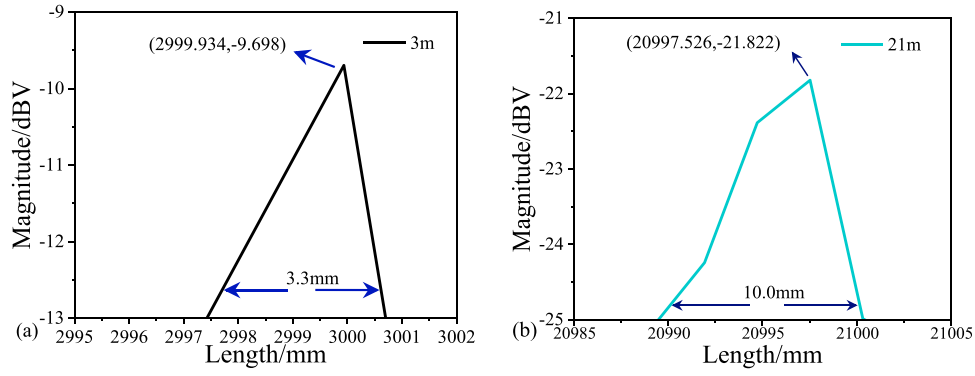


Fig. 8. (a) 3 m system resolution. (b) 21 m system resolution.

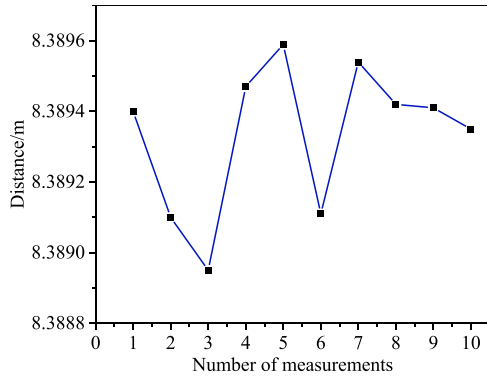


Fig. 9. Repeated ranging results in free space.

One of the reasons why less than 20 % of the remaining frequencies are not locked is the hysteresis effect of the DFB laser, in which the output laser frequency is out of sync with the driving current, the other reason is that the input current of DFB laser is at the turning point, the temperature of the laser changes suddenly, causing the beat signal to lose Lock-in.

Analyzing the three stages of the DFB correction process and compared with no correction, the HEO-PLL proposed in this paper sacrifices the laser frequency sweeping rate of the DFB laser, and the center frequency of the beat frequency signal decreases by 69.5 kHz, and the sweeping rate decreases by 2.772 THz/s. However, it brings a more stable and higher linearity DFB laser frequency sweeping. The nonlinearity of the DFB laser frequency sweeping decreases from 16.55 % to 0.078 %, and the power of the center frequency of the beat frequency signal increases by 31.49 dB.

3.3. Ranging experiment

To further verify the correction results of the HEO-PLL, this paper conducted an experiment of fiber length measurement under the

combination of open-loop soft correction and HEO-PLL of the DFB laser. Fig. 7(a) shows the measurement results and fitting curves for different lengths of optical fibers, and Fig. 7(b) shows the FFT power spectrum of the beat frequency signal generated by the DFB laser under the correction system with different lengths of optical fibers. It can be observed that the beat frequency signal can still be resolved under the 21 m test fiber. Fig. 7(b) shows the measurement results and fitting curves for different lengths of optical fibers. The results of fitting curve show that the linearity between beat signal and measured fiber length is 1, and the residual sum of squares is about 8.73×10^{-5} . The maximum error of the system is 8 mm in the range of 210m.

In the fiber length test experiment, the 3 m and 21 m test results are shown in Fig. 8. We can find that the resolution of the system is 3.3 mm at the 3 m test distance and the resolution of the system is reduced to 10.0 mm at the 21 m test distance. The result of the free-space distance detection experiment is shown in Fig. 9, we carried out ten experiments and got the standard deviation of 210um.

Finally, we compare this method with other linear frequency scanning methods in Table 1. The results show that the HEO-PLL method can realize high frequency linear scanning under high frequency scanning slope, and it allows real-time feedback and self-regulation in the experimental process, avoiding external interference. Although the experiment is based on the measurement of fiber length, it has some advantages in the measurement accuracy of the distance measurement system using DFB laser.

4. Conclusion

In this paper, we have demonstrated that the open-loop soft correction combined with the HEO-PLL can be used to linearize the frequency sweep of a laser, which can be applied to FMCW LiDAR and easily adapted to similar linear frequency sweeps of current-tuned DFB lasers. This method reduces the impact of external temperature and driving signal fluctuations on the laser through the software and hardware feedback, making it easy to adjust. In the experiment, we achieved a triangular wave sweep with a sweep rate of 6.78 THz/s and nonlinearity of 0.078 %, and the power of the beat signal was increased by 31.49 dB.

Table 1
Comparison of ranging results of different DFB laser correction systems.

Methods	Open/closed loop	Whether the beat signal is processed by algorithm	Maximum range	free space/fiber	Precision	Resolution
This work	Open and closed	No	21.00m	Fiber	8mm	10mm
Correction algorithm [25]	Open	No	16.00m	Free space	50mm	/
A nonlinearity-compensation-free (OFDR) scheme [13]	Open	No	2110m	Fiber	/	26.7cm
Electronic-photonic phase-locked loop [26]	Closed	Yes	1.40m	Free space	50mm	8μm
Digital electro-optical phase-locked loop [19]	Closed	Yes	10.3m	Fiber	/	16.35μm
A phase spread frequency sampling [17]	Open	Yes	1.64m	Free space	/	19μm

In addition, we conducted fiber length measurements using the corrected laser and achieved a maximum error of 8 mm and a minimum resolution of 10 mm within a range of 21 m.

CRedit authorship contribution statement

Jiatong Zhang: Methodology, Writing – original draft. **Chang Liu:** Data curation, Conceptualization. **Liwen Su:** Formal analysis, Validation. **Xinghu Fu:** Writing – review & editing. **Wa Jin:** Investigation. **Weihong Bi:** Resources, Visualization. **Guangwei Fu:** Supervision, Project administration, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgment

This work was supported by the Natural Science Foundation of Hebei Province (F2021203045, F2021203058); the Hebei Province Science and Technology Support Program (216Z1702G, 216Z1706G); and the National Natural Science Foundation of China (61575170).

References

- [1] Riemensberger J, Lukashchuk A, Karpov M, Weng W, Lucas E, Liu J, Kippenberg TJ. Massively parallel coherent laser ranging using a soliton microcomb. *Nature* 2020;581(7807):164–70. <https://doi.org/10.1038/s41586-020-2239-3>.
- [2] Sun C, Chen Z, Ye S, Lin J, Shi W, Li B, Teng F, Li X, Zhang A. Highly-time-resolved FMCW LiDAR with synchronously-nonlinearly-corrected acquisition for dynamic locomotion. *Opt Express* 2023;31(5):7774–88. <https://doi.org/10.1364/OE.480346>.
- [3] Zhang X, Pouls J, Wu MC. Laser frequency sweep linearization by iterative learning pre-distortion for FMCW LiDAR. *Opt Express* 2019;27(7):9965–74. <https://doi.org/10.1364/OE.27.009965>.
- [4] Kim C, Jung Y, Lee S. FMCW LiDAR system to reduce hardware complexity and post-processing techniques to improve distance resolution. *Sensors* 2020;20(22):6676. <https://doi.org/10.3390/s20226676>.
- [5] Zhang F, Yi L, Qu X. Simultaneous measurements of velocity and distance via a dual-path FMCW lidar system. *Opt Commun* 2020;474:126066. <https://doi.org/10.1016/j.optcom.2020.126066>.
- [6] Lu C, Yu Z, Liu G. A high-precision range extraction method using an FM nonlinear kernel function for DFB-array-based FMCW lidar. *Opt Commun* 2022;504:127469. <https://doi.org/10.1016/j.optcom.2021.127469>.
- [7] Suyama S, Ito H, Kurahashi R, Abe H, Baba T. Doppler velocimeter and vibrometer FMCW LiDAR with Si photonic crystal beam scanner. *Opt Express* 2021;29(19):30727. <https://doi.org/10.1364/OE.438453>.
- [8] Zhang J, Gao C, Xue M, Liu R. Research on frequency modulation character of the current driven DFB semiconductor laser. *Mod Phys Lett B* 2019;33(1):1850422. <https://doi.org/10.1142/s0217984918504225>.
- [9] Hyun E, Jin Y. Doppler-spectrum feature-based human-vehicle classification scheme using machine learning for an FMCW radar sensor. *Sensors* 2020;20(7):2001. <https://doi.org/10.3390/s20072001>.
- [10] Mingshi Z, Yubing W, Lanxuan Z, Qian H, Shuhua Z, Lei L, Yongyi C, Li Q, Junfeng S, Lijun W. Phase-modulated continuous-wave coherent ranging method for optical phased array lidar. *Opt Express* 2023;31(4):6514–28. <https://doi.org/10.1364/OE.27.013160>.
- [11] Lou X, Yuan Z, Dong Y. Rapid spectroscopic gas sensing using optical linear chirp chain. *Opt Express* 2019;27(9):13160–71. <https://doi.org/10.1364/OE.27.013160>.
- [12] Cheng X, Liu J, Zhang Y, Zhang F, Qu X. Simultaneous measurement of distance and speed via frequency-modulated continuous-wave LiDAR system based on H13C14N gas cell. *Opt Lasers Eng* 2022;159:107193. <https://doi.org/10.1016/j.optlaseng.2022.107193>.
- [13] Yang F, Zhang LJ, Zhang ZY, Zhou XJ, Liu Y. Nonlinearity-compensation-free optical frequency domain reflectometry based on electrically-controlled optical frequency sweep. *J Electron Sci Technol* 2021;19(1):100025. <https://doi.org/10.1016/j.jnlest.2020.100025>.
- [14] Na Q, Xie Q, Zhang N, Zhang L, Li Y, Chen B, Peng T, Zuo G, Zhuang D, Song J. Optical frequency shifted FMCW Lidar system for unambiguous measurement of distance and velocity. *Opt Lasers Eng* 2023;164:107523. <https://doi.org/10.1016/j.optlaseng.2023.107523>.
- [15] Gao S, Luo M, Jing Z, Chen H. A tunable dual-wavelength fiber ring-cavity laser based on a FBG and DFB laser injection. *Optik* 2020;203:163961. <https://doi.org/10.1016/j.ijleo.2019.163961> (Stuttg).
- [16] Ye S, Che M, Kuboki T, Kato K. Fast wavelength switching at tunable DFB laser array by current/temperature cooperative control. *IEEE Photonics Technol Lett* 2022;34(1):51–4. <https://doi.org/10.1109/lpt.2021.3137808>.
- [17] Zhang X, Kong M, Guo T, Zhao J, Wang D, Liu L, Liu W, Xu X. Frequency modulation nonlinear correction and range-extension method based on laser frequency scanning interference. *Appl Opt* 2021;60(12):3446–51. <https://doi.org/10.1364/AO.420663>.
- [18] X. Mao, X. Ji, Y. Tan, H. Ye, X. Wang, High-sensitivity all-optical PA spectrometer based on fast swept laser interferometry, *Photoacoustics*, 2022, 28: 100391. <https://doi.org/10.1016/j.pacs.2022.100391>.
- [19] Hauser M, Hofbauer M. FPGA-based EO-PLL with repetitive control for highly linear laser frequency tuning in FMCW LiDAR applications. *IEEE Photonics J* 2022;14(1):6808608. <https://doi.org/10.1109/jphot.2021.3139053>.
- [20] Yao Z, Xu Z, Mauldin T, Heffernan G, Wei T. A reconfigurable architecture for continuous double-sided swept-laser linearization. *J Lightwave Technol* 2020;38(18):5170–6. <https://doi.org/10.1109/jlt.2020.2999825>.
- [21] Satyan N, Vasilyev A, Rakuljic G, Leyva V, Yariv A. Precise control of broadband frequency chirps using optoelectronic feedback. *Opt Express* 2009;17(18):15991–9. <https://doi.org/10.1364/OE.17.015991>.
- [22] Wang Y, Xie W, Chen H, Day-Wei Li D. High-resolution time-to-digital converters (TDCs) with a bidirectional encoder. *Measurement* 2023;206:112258. <https://doi.org/10.1016/j.measurement.2022.112258>.
- [23] Cao X, Wu K, Li C, Zhang G, Chen J. Highly efficient iteration algorithm for a linear frequency-sweep distributed feedback laser in frequency-modulated continuous wave lidar applications. *J Opt Soc Am B* 2021;38(10):D8. <https://doi.org/10.1364/josab.430605>.
- [24] Li P, Zhang Y, Yao J. Linear frequency swept laser source with high swept slope based on digital optical phase-locked loop. *Opt Commun* 2022;525:128860. <https://doi.org/10.1016/j.optcom.2022.128860>.
- [25] Cao X, Song P, Pan Z, Liu B. Correction algorithm of the frequency-modulated continuous-wave LiDAR ranging system. *Opt Express* 2021;29(21):34150–65. <https://doi.org/10.1364/OE.438158>.
- [26] Behroozpour B, Sandborn PAM, Quack N, Seok TJ, Matsui Y, Wu MC, Boser BE. Electronic-photonic integrated circuit for 3D microimaging. *IEEE J Solid State Circuits* 2017;52(1):161–72. <https://doi.org/10.1109/jssc.2016.2621755>.